HDOC- 4 – 1A

NAME : DARSHIL RAJ

DATE : 2/10/26 **SAP ID:** 590040196

Program to Swap Two Numbers Without Using a Third Variable

This document provides a clean, well-commented implementation for swapping two numbers without using a third variable in C. The program takes two numbers as input and exchanges their values using arithmetic operations.

1. C Implementation

Step 1: Understand the Logic

• The two numbers are swapped without using any additional variable.
• Arithmetic operations are used to exchange the values of the two numbers.

Step 2: Define Variables and Data Types

• Use integer variables (int) to store the two numbers.

Step 3: Take User Input

• Read two numbers using scanf("%d %d", &num1;, &num2;).

Step 4: Swap the Values

1. Compute num1 = num1 + num2.
2. Compute num2 = num1 - num2.
3. Compute num1 = num1 - num2.

Step 5: Format the Output

• Print the values of the two numbers before and after swapping.

Complete C Code

```
/*
 * Q4 (User Inputs, Operations & Output)
 * Write a program to swap two numbers without using a third variable.
 */

#include <stdio.h>

int main() {
    int num1, num2;

    // Step 1: Read two numbers
    scanf("%d %d", &num1, &num2);

    // Step 2: Display values before swapping
    printf("Before swapping: num1=%d, num2=%d\n", num1, num2);

    // Step 3: Swap without using a third variable
    num1 = num1 + num2;
    num2 = num1 - num2;
    num1 = num1 - num2;

    // Step 4: Display values after swapping
    printf("After swapping: num1=%d, num2=%d\n", num1, num2);
```

```
    return 0;
}
```

Dry Run with Test Cases

Sample Test Case 1:

• Input: 10 20
• Calculation:
o num1 = 10 + 20 = 30
o num2 = 30 - 20 = 10
o num1 = 30 - 10 = 20
• Output: Before swapping: num1=10, num2=20
After swapping: num1=20, num2=10

Sample Test Case 2:

• Input: 5 15
• Calculation:
o num1 = 5 + 15 = 20
o num2 = 20 - 15 = 5
o num1 = 20 - 5 = 15
• Output: Before swapping: num1=5, num2=15
After swapping: num1=15, num2=5